To whom it may concern,

As the chief mentor of an AI + hardware research program, I am delighted to write this letter of recommendation for my student, Chengji Song. In January 2025, Chengji Song participated in this short-term research program, which significantly enhanced his comprehension of the "theory-technology-product" tripartite AI system development model and improved his practical hardware-software integration capabilities. As his primary academic supervisor, I have witnessed his exceptional academic performance and remarkable research ability as well as his profound interest and potential in artificial intelligence and autonomous driving technologies.

During the project, Chengji Song systematically researched three representative applications of AI technologies in the field of autonomous driving: visual perception, path planning, and cruise control. Besides, he also successfully completed five critical experiments, including multi-sensor fusion, YOLO-based image recognition, SLAM-based localization and mapping, path planning algorithm development, and PCC (Predictive Cruise Control) simulation validation, with the aid of MATLAB, ROS, TruckSim and other software. Finally, he successfully realized the real-vehicle deployment, mapping and navigation of Mecanum wheel intelligent vehicles and well mastered relevant basic theories and core technologies.

Based on the theoretical foundation of the project, Chengji Song determined a more in-depth and specific research topic titled "Application of Autonomous Driving in Unmanned Trucks". During the research process, Chengji Song not only participated in the overall research but also conducted a relatively in-depth study on multi-sensor fusion and the Kalman filter algorithm. He summarized the principles and methods of data-level fusion, feature-level fusion, and decision-level fusion, contributing to the data processing part of the subject. At the same time, he showed extremely strong enthusiasm and was very good at drawing inferences about other cases from one instance. He could analyze potential risks according to research contents and specific application scenarios and propose solutions, which provided solid support for the successful completion of research tasks. Besides, based on the SLAM technology, he completed the positioning of the Mecanum wheel intelligent vehicle and the construction of a real-scene map. In addition, he carefully analyzed advantages and disadvantages of the Dijkstra algorithm and the A* algorithm, and finally selected Dijkstra algorithm as the optimal path planning solution, integrating working principles and specific scenarios of PCC (Predictive Cruise Control). Finally, with the aid of the high-speed road map of China and the road map of Beijing, he clearly expounded the application status, advantages, and prospects of autonomous driving in unmanned trucks and successfully completed research and presentation tasks of the topic.

In conclusion, I deeply appreciate Chengji Song's research spirit, excellent practical ability and his heartfelt love for artificial intelligence technology. I believe that he has possessed all excellent comprehensive qualities necessary for scientific research in the field of artificial intelligence. I hope this letter can provide some valuable reference for you to evaluate his academic performance. Please feel free to contact me if you have any questions about this project.

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